

LIBRARY MANAGEMENT SYSTEM

Using MongoDB

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1. Introduction

The Library Management System is a database project developed using MongoDB. It helps manage books, students or members, issue records and return records. MongoDB stores information in flexible document format, making it suitable for this project.

2. Objectives

The main objectives are to store library information, maintain book records, manage members, record issued and returned books, search records quickly, and perform CRUD operations using MongoDB.

3. Technologies Used

MongoDB, MongoDB Shell/Compass, JSON/BSON documents, and basic database query commands are used.

4. System Requirements

Hardware: basic computer/laptop with at least 4 GB RAM. Software: MongoDB Community Server or MongoDB Atlas, MongoDB Compass, and a modern web browser.

5. MongoDB Introduction

MongoDB is a NoSQL document-oriented database. Instead of rows and columns, it stores data as documents inside collections. Documents are similar to JSON objects and are stored internally in BSON format.

6. Database Design

Database name: libraryDB. Collections: books, members, and issueRecords.

7. Collections and Fields

Books: bookId, title, author, category, quantity. Members: memberId, name, course, phone. IssueRecords: issueId, bookId, memberId, issueDate, returnDate, status.

8. Create Database and Collections

MongoDB creates a database when data is first stored in it. The following commands select the database and create collections.

9. Insert Sample Data

Sample records can be inserted into books, members and issueRecords collections.

10. CRUD Operations

CRUD means Create, Read, Update and Delete. MongoDB provides insertOne/insertMany, find, updateOne/updateMany and deleteOne/deleteMany for these operations.

11. Useful Queries

The project can search books by title, find available books, display members, update book quantity, and remove old records.

12. Advantages

Easy to use, flexible document structure, scalable, fast for many common operations, and suitable for applications where data structure may change.

13. Limitations

Complex relational operations can be less convenient than in relational databases, and careful database design is required for large systems.

14. Conclusion

The Library Management System demonstrates how MongoDB can be used to store and manage library data. It covers database creation, collections, documents and CRUD operations. The project provides a simple foundation that can later be connected to a Python, Java or web application.

15. MongoDB Commands / Practical

```
use libraryDB
db.createCollection("books")
db.createCollection("members")
db.createCollection("issueRecords")

db.books.insertMany([
  {bookId:1,title:"Python Basics",author:"R. Sharma",category:"Programming",quantity:5},
  {bookId:2,title:"Database Management",author:"A. Kumar",category:"DBMS",quantity:3},
  {bookId:3,title:"Computer Networks",author:"S. Gupta",category:"Networking",quantity:4}
])

db.members.insertMany([
  {memberId:101,name:"Sapna Rao",course:"BCA",phone:"9876543210"},
  {memberId:102,name:"Ananya Singh",course:"BCA",phone:"9876543211"}
])

db.issueRecords.insertOne({
  issueId:1001, bookId:1, memberId:101,
  issueDate:"2026-10-06", returnDate:null, status:"Issued"
})

db.books.find()
db.books.find({category:"Programming"})
db.books.updateOne({bookId:1}, {$inc:{quantity:-1}})
db.members.find({course:"BCA"})
db.issueRecords.find({status:"Issued"})
db.members.deleteOne({memberId:102})
```

16. Expected Output

The commands display stored books and members, search records according to conditions, update quantities when a book is issued, and delete selected member records. MongoDB Compass can be used to view the documents graphically.